

Harish Balasubramanian

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PROFESSIONAL SUMMARY

Biomedical engineer bridging **fluorescence microscopy instrumentation** and **AI-driven biological image analysis** to accelerate discovery. Builds tools that let researchers see deeper, quantify faster, and draw conclusions that were previously inaccessible across live-cell imaging, volumetric reconstruction, and physiological signal processing.

EDUCATION

Johns Hopkins University

Baltimore, MD

Master of Science in Engineering, Biomedical Engineering (GPA: 3.87/4)

Aug 2024 – Present

- **Relevant Coursework:** Machine Learning for Signal Processing, Physics of Medical Imaging, **Deep Learning**, Design of Advanced Biomedical Instruments, **Machine Intelligence for Embedded Systems**

Sardar Patel Institute of Technology

Mumbai, India

Bachelor of Technology, Electronics and Telecommunications Engineering (GPA: 3.6/4.0)

Sept 2020 – May 2024

- **Thesis:** AI-driven UAV-ground robot system for precision agriculture; awarded a **\$25,000** IEEE AESS DSTEI Global Grant

EXPERIENCE

AI Intern

Feb 2026 – Present

HDMedical Group

Santa Clara, CA

- Trained **BiLSTM** model achieving **99% AFib** and **89% murmur** detection on HD Steth, enabling reliable cardiac screening on a portable device accessible outside hospital settings
- Adapted **ECG-FM** (wav2vec 2.0) to a **7-lead handheld ECG+PCG device** via **random lead masking**, bringing hospital-grade arrhythmia classification to point-of-care use with **LLM-generated clinical reports**, versioned via **CI/CD**

Research Assistant

Oct 2024 - Present

Dr. Pathak Lab, Johns Hopkins School of Medicine

Baltimore, MD

- Trained **Pix2Pix-GAN** and **SwinIR** models for super-resolved cortical denoising (**44dB PSNR**, **99.2% SSIM**), eliminating need for high-intensity illumination and enabling long-term imaging in freely behaving mice without phototoxicity
- Demonstrated that denoised stacks fully preserve **functional connectivity** and vascular dynamics without temporal input, validating the approach as a method for low-light in-vivo cortical imaging compatible with miniscope systems
- Built a wireless **Raspberry Pi** fluorescence microscope with **EWL** autofocus and CAD-modeled optomechanics, enabling continuous incubator-safe imaging of tumor cells and **brain organoids** without disrupting sterility or cell viability. Controlled via I2C-linked **MAX14574** driver and remote GUI
- Fine-tuned **CellPose** and **DeepWonder** for automated cell segmentation and tracking, enabling quantification of tumor dynamics across **temporal in-vivo** stacks that would be infeasible to annotate manually

Teaching Assistant

Aug 2025 – Dec 2025

Johns Hopkins University

Baltimore, MD

- TA for Biomedical Instrumentation (Dr. Thakor), guiding **30+ students** through analog front-end design and biosignal acquisition to build functional ECG, EMG, and PPG systems
- TA for AI Systems (Dr. Dredze), coaching teams on **deployable AI** with emphasis on human-centered design

Biomedical Engineering Intern

May 2025 - August 2025

Xenco Medical

San Diego, CA

- Built a first-of-its-kind **Cellular Potts Model** of multicellular bone regeneration, replicating 12-week in-vivo graft trends with **95% agreement**, reducing the need for costly animal experiments in scaffold development
- Mapped porosity, osteoconductivity, and resorption rate to tissue outcomes, enabling **in-silico** screening of graft formulations

Research Intern

Jan 2023 – Jun 2023

Indian Institute of Technology, Bombay

Mumbai, India

- Designed Electric impedance tomography **PCB** and trained **Radial basis autoencoder model** for inverse conductivity reconstruction in lung phantoms as a low-cost, radiation-free alternative to CT for bedside pulmonary monitoring

SKILLS

Programming & ML: Python, C++, C, Java, MATLAB, TensorFlow, PyTorch, OpenCV, TensorRT, TFLite, TinyML

Biomedical AI & Signal/Image Analysis: physiological signal analysis (ECG, PCG, EMG, PPG), biosignal acquisition, analog front-end design, medical image segmentation, 3D/4D volumetric imaging, instance segmentation, ImageJ/Fiji, DICOM, MIPAV

Embedded Systems & Hardware: Raspberry Pi, Arduino, STM32, KiCad, Altium Designer, PCB prototyping, I2C, register-level device control, driver bring-up, electrowetting optics, embedded AI deployment, real-time system design, LiDAR-based navigation, SOLIDWORKS

Tools, Platforms & Process: Git, Docker, CI/CD (GitHub Actions), VS Code, PyCharm, Google Cloud Platform, AWS, software documentation, design verification, quality systems awareness, quantitative validation, model benchmarking, NCSS

PUBLICATIONS

Deep Learning Preserves Functional Connectivity in Low-Light In-Vivo Cortical Imaging	2025
<i>Alavi-Bradley Symposium on Molecular Imaging and Theranostics</i>	Link
ML-based Detection of Covid-19, Pneumonia, and Tuberculosis from Chest X-Rays	2023
<i>arXiv Preprint</i>	Link

PROJECTS

Medhive.ai – AI/Knowledge Graph Prototyping	Apr 2026 – Present
<i>Medhive.ai</i>	
- Exploring knowledge graph based representations (Neo4j) for structuring medical device and regulatory data into queryable, reasoning-capable systems - Prototyping GraphRAG for reasoning in LLMs for contextual retrieval and early-stage regulatory decision support	
CoREAM SegNet	Jan 2025 – May 2025
<i>Johns Hopkins University</i>	<i>Dr. Vishal Patel</i>
Designed a compact CNN–Attention hybrid with multi-scale feature fusion for real-time medical image segmentation, achieving 2% higher DICE and mIoU than transformer models on EndoVis2018, LiTS2017, and BraTS2020 medical datasets while using a fraction of the compute, making clinical deployment viable on constrained hardware	
Multimodal Acoustic–Ultrasonic Gesture Recognition on Edge Hardware	Jan 2025 – May 2025
<i>Johns Hopkins University</i>	<i>Dr. Andreas Andreou</i>
Built a dual-modality gesture system fusing passive acoustic (25 kHz) and active ultrasonic (40–50 kHz) reflections via multi-head attention CNN; quantized to 94 kB TFLite for Arduino Nano, enabling touchless control in sterile or busy environments at the edge	
High Density Tactile Glove	Sept 2024 - Dec 2024
<i>Neuroengineering and Biomedical Instrumentation Lab, Johns Hopkins University</i>	<i>Dr. Nitish Thakor</i>
Designed a two-layer flex-PCB glove with 1024 tactile sensors in KiCad using compressed sensing, providing high-density haptic feedback at low latency for prosthetic and wearable applications	
Assistive Biogaming — EOG Minecraft Controller	Sept 2024 – Dec 2024
<i>Johns Hopkins University</i>	<i>Dr. Nitish Thakor</i>
Designed a custom analog front-end (INA128, HPF/LPF chain, gain 605) for electrooculography and implemented embedded C++ gesture detection on Arduino for fully hands-free Minecraft control for users with motor impairments	
Drone for Smart Agriculture	Sept 2023 – May 2024
<i>IEEE AESS Global DSTEI Project</i>	Grant Link
Led AI-enabled UAV–ground robot system for precision agriculture, securing a \$25,000 IEEE AESS DSTEI grant; built 2 kg & 5 kg prototypes (T/W > 2.3) with A*+RRT LiDAR navigation for targeted autonomous pesticide delivery, reducing chemical waste and field labor	